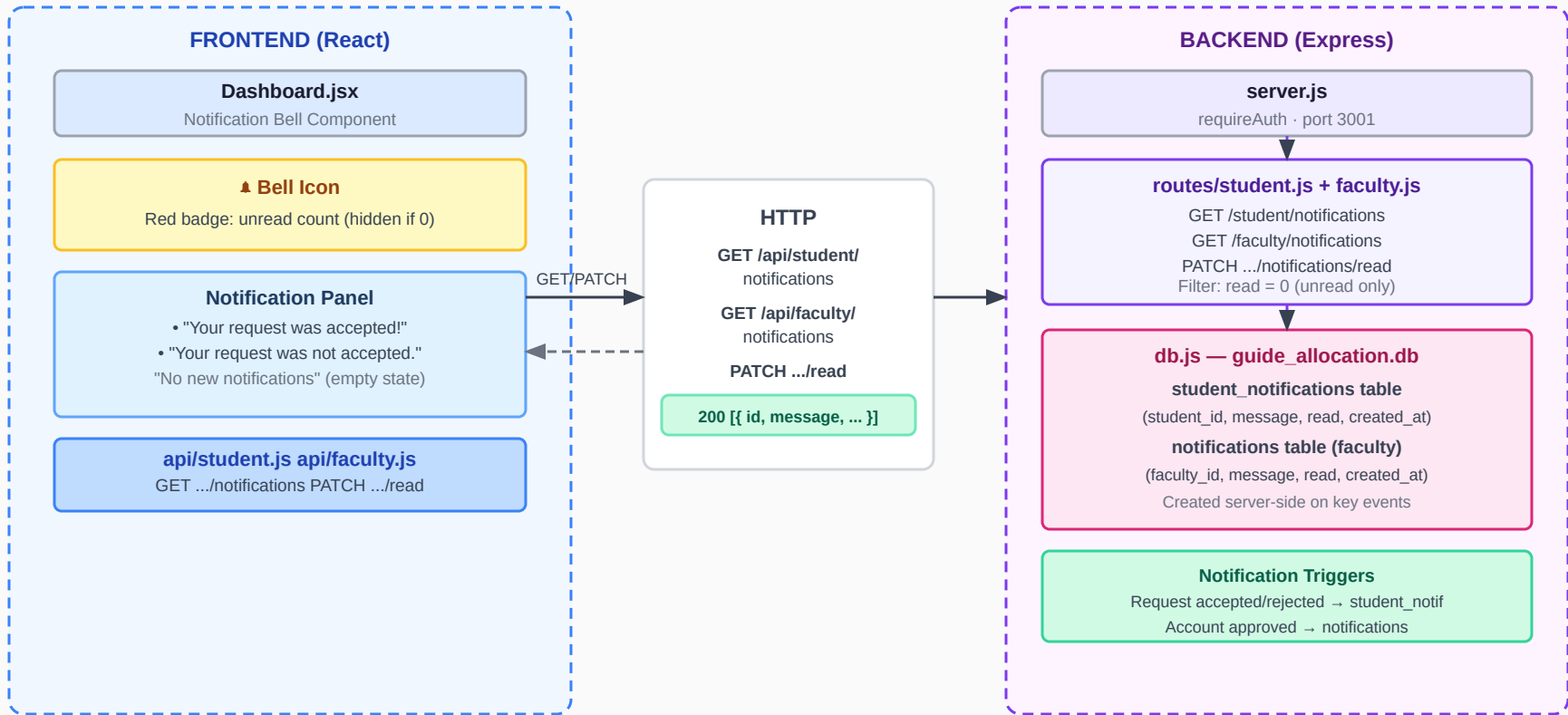


US9 — Notification System

Architecture Wireframe



US9 — Notification System

Document ID	GAS-FD-US9	Version	1.0
Prepared By	Development Team	Date	April 2025
User Story	As a user, I want to receive in-app notifications so that I stay updated on approval and allocation status.		

1. User Story

US9 — Notification System

"As a user, I want to receive in-app notifications so that I stay updated on approval and allocation status."

Acceptance Criteria:

- Given a status change (request accepted, rejected, or account approved), when it occurs, then the affected user should receive an in-app notification.
- Given unread notifications exist, when the user views the dashboard, then the bell icon should display an unread count badge.

2. Functional Description

A notification bell icon on the dashboard header shows a red badge with the count of unread notifications. Notifications are created server-side when key system events occur: a student's guide request is accepted or rejected, a faculty member's account is approved by the admin, or a faculty member receives a new student request. On the dashboard, the bell click reveals the notification panel. Clicking 'Mark all as read' or any read action clears the badge and marks records in the database.

3. Notification Triggers

Event	Recipient	Message Template	Table
Faculty accepts student request	Student	Your guide request to Prof. [name] was accepted!	student_notifications

Event	Recipient	Message Template	Table
Faculty rejects student request	Student	Your guide request to Prof. [name] was not accepted.	student_notifications
Auto-cascade rejection (on another acceptance)	Student	Your request to Prof. [name] was automatically rejected since you have been assigned a guide.	student_notifications
Admin approves faculty account	Faculty	Your account has been approved. You can now log in.	notifications
Student sends guide request	Faculty	(Implicit — faculty sees request in pending dashboard; no separate notification row in current implementation)	—

4. Business Rules

- Notifications are created server-side only — no client-side fabrication.
- Student notifications use the `student_notifications` table; faculty notifications use the `notifications` table.
- Only **unread** notifications are returned by the GET endpoints (`read = 0`).
- Marking as read is done via `PATCH .../notifications/read` — bulk operation (marks all unread as read).
- Notifications are user-scoped: each user only receives their own notifications.
- No email notifications in v1 — in-app only.
- Notification records are never deleted; they are only updated to `read = 1` .

5. Process Flow

#	Step	Actor	Details
1	System event occurs	Backend	e.g., faculty accepts request via <code>PATCH /api/requests/:id/status</code>
2	Notification created	Backend / DB	<code>INSERT INTO student_notifications (student_id, message) VALUES (?, ?)</code>
3	Student logs in / loads dashboard	Frontend	Dashboard.jsx mounts

#	Step	Actor	Details
4	Dashboard fetches notifications	Frontend	GET /api/student/notifications — returns unread rows
5	Badge renders	Frontend	Bell icon shows red badge with count; hidden if count = 0
6	User clicks bell / 'Mark all read'	Student	Triggers PATCH /api/student/notifications/read
7	Backend marks read	Backend / DB	UPDATE student_notifications SET read = 1 WHERE student_id = ?
8	Badge clears	Frontend	Re-fetch returns empty array; badge hidden

6. API Specification

Student Notifications

Property	Value
GET URL	/api/student/notifications
Auth	Bearer JWT (student role)
200 — Success	[{ id, message, created_at }, ...] — unread notifications only
403	{ "error": "Forbidden" } — non-student token

Mark Student Notifications Read

Property	Value
PATCH URL	/api/student/notifications/read
Auth	Bearer JWT (student role)
200 — Success	{ "message": "Marked as read" }

Faculty Notifications

Property	Value
GET URL	/api/faculty/notifications
Auth	Bearer JWT (faculty role)

Property	Value
200 — Success	<code>[{ id, message, created_at }, ...]</code> — unread notifications only

Mark Faculty Notifications Read

Property	Value
PATCH URL	<code>/api/faculty/notifications/read</code>
Auth	Bearer JWT (faculty role)
200 — Success	<code>{ "message": "Marked as read" }</code>

7. UI Behaviour

Scenario	UI Response
Unread notifications exist	Bell icon shows a red badge with count number
No unread notifications	Bell icon shown without badge
User clicks 'Mark all read'	PATCH sent; badge disappears; notification items styled as read
Notification panel empty	'No new notifications' placeholder text shown
Network error fetching notifications	Bell shown without badge; silent fail (non-critical path)

8. Database Schema

student_notifications table

Column	Type	Constraint
id	INTEGER	PRIMARY KEY AUTOINCREMENT
student_id	INTEGER	NOT NULL, FOREIGN KEY → students(id) ON DELETE CASCADE
message	TEXT	NOT NULL
read	INTEGER	DEFAULT 0
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

notifications table (faculty)

Column	Type	Constraint
id	INTEGER	PRIMARY KEY AUTOINCREMENT
faculty_id	INTEGER	NOT NULL, FOREIGN KEY → faculty(id) ON DELETE CASCADE
message	TEXT	NOT NULL
read	INTEGER	DEFAULT 0
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP

US9 — Notification System

Document ID	GAS-TC-US9	Version	1.0
Prepared By	Development Team	Date	April 2025
Environment	Frontend: localhost:5173 · Backend: localhost:3001 · DB: SQLite (guide_allocation.db)		
Precondition	Both servers running; student and faculty accounts exist; at least one accepted/rejected request in DB		

Test Summary

6 Total	6 Pass	0 Fail	0 Pending
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Test Cases

TC ID	Test Case	Priority	Preconditions	Test Steps	Expected Result	Status
TC-US9-01	Student receives notification when request accepted	High	Faculty has just accepted student's guide request	1. Login as the student 2. Check bell icon on dashboard	Bell shows badge count ≥ 1 ; notification message: 'Your guide request to Prof. [name] was accepted!'	PASS
TC-US9-02	Student receives notification	High	Faculty has rejected	1. Login as the student 2. Check	Bell shows badge; notification	PASS

TC ID	Test Case	Priority	Preconditions	Test Steps	Expected Result	Status
	when request rejected		student's guide request	bell icon on dashboard	message: 'Your guide request to Prof. [name] was not accepted.'	
TC-US9-03	Student receives auto-rejection notifications on cascade	High	Student had pending requests to 3 faculty; one accepted — other 2 auto-rejected	1. Login as student after one acceptance 2. Check notification panel	3 notifications: 1 acceptance + 2 auto-rejection messages (one per auto-rejected faculty)	PASS
TC-US9-04	Faculty receives notification when account approved	High	Admin has just approved a faculty account	1. Login as the newly approved faculty 2. Check bell icon	Bell shows badge; notification: 'Your account has been approved. You can now log in.'	PASS
TC-US9-05	Bell badge clears after marking notifications as read	High	Student has unread notifications	1. Login as student with unread notifications 2. Click 'Mark all read'	Badge disappears; GET /api/student/notifications returns empty array	PASS
TC-US9-06	User sees only their own notifications	Medium	Student A and Student B both have notifications	1. Login as Student A 2. Inspect notification list 3. Verify Student B's	Only Student A's notifications shown; Student B's messages absent	PASS

TC ID	Test Case	Priority	Preconditions	Test Steps	Expected Result	Status
				messages not present		